

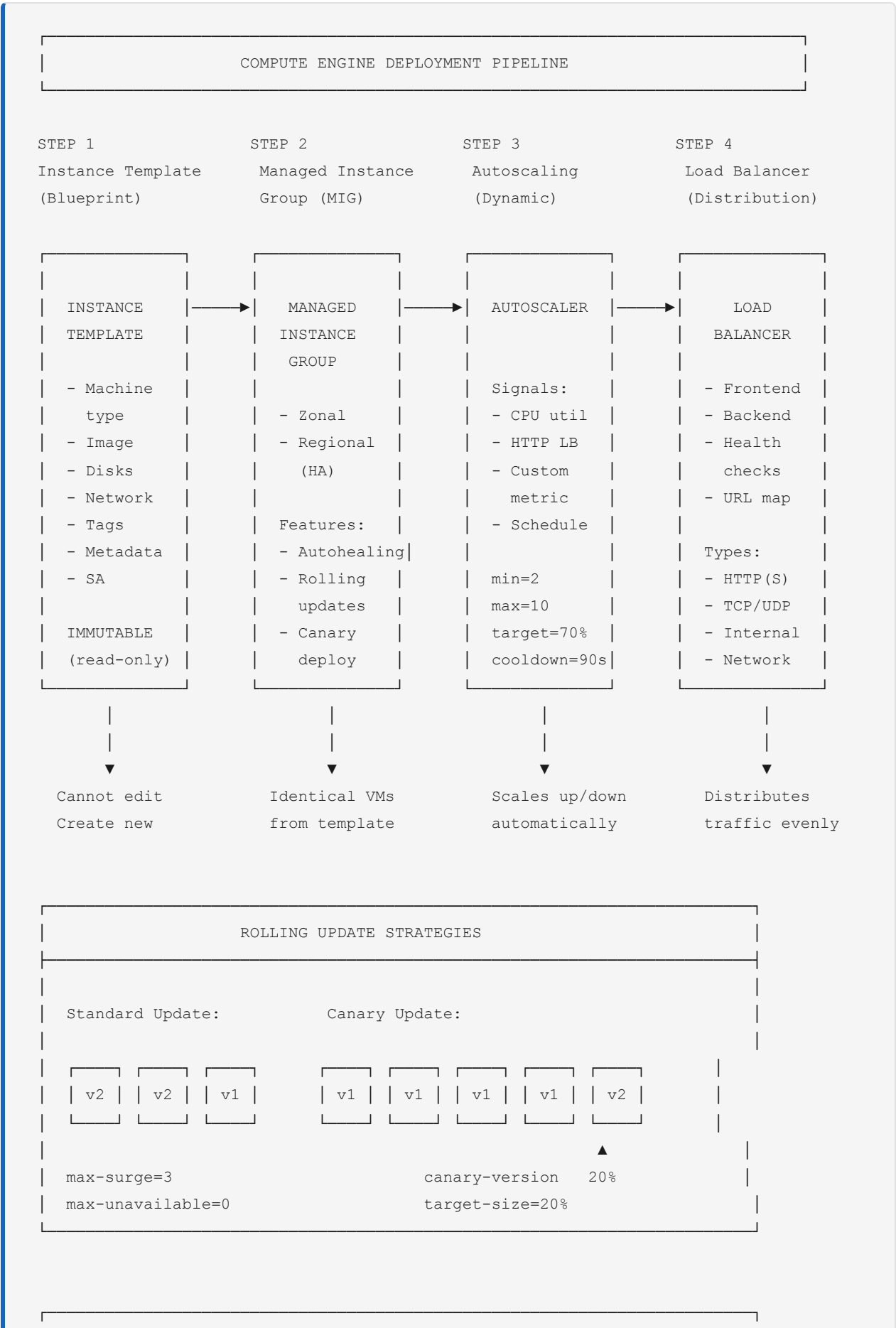
GOOGLE CLOUD PROFESSIONAL CLOUD ARCHITECT

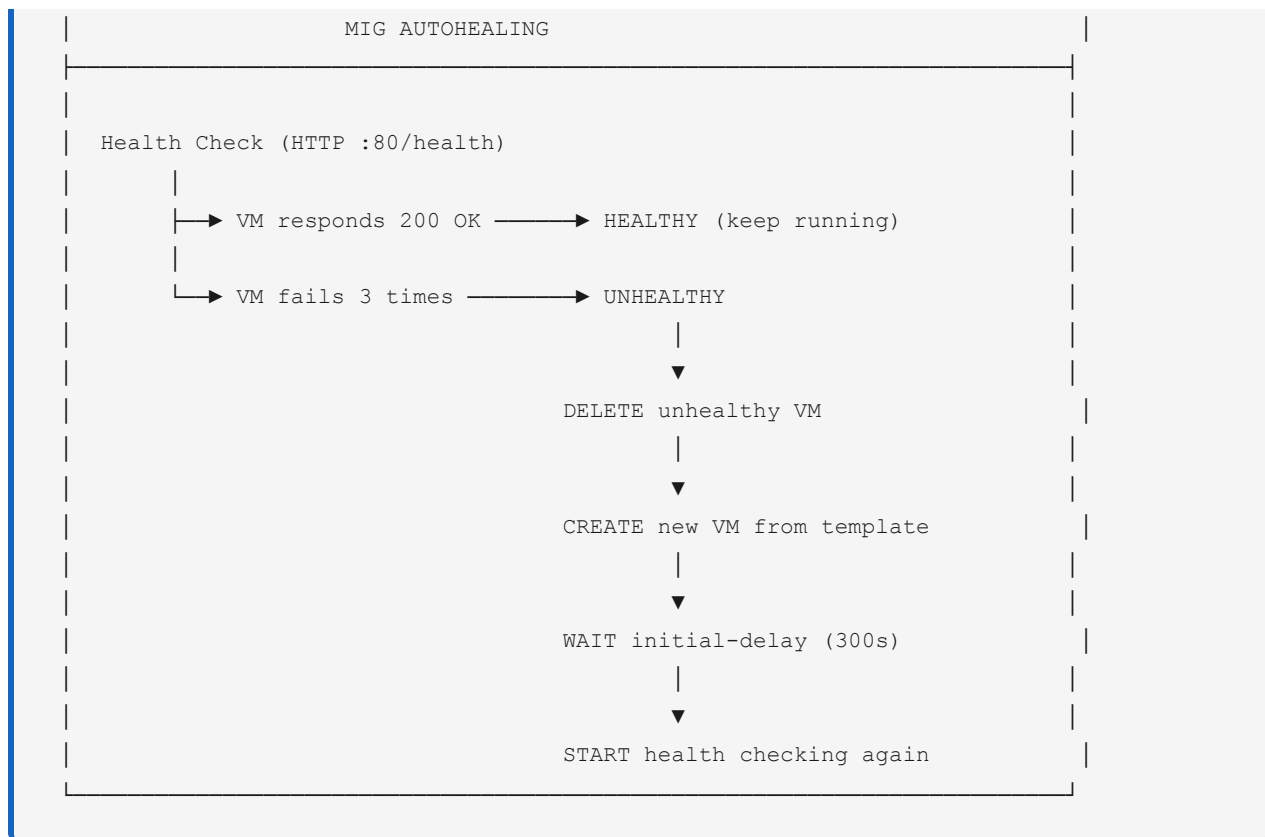
Section 3 -- Deploying and Implementing Cloud Solutions -- Visual Maps

GCP Study Notes

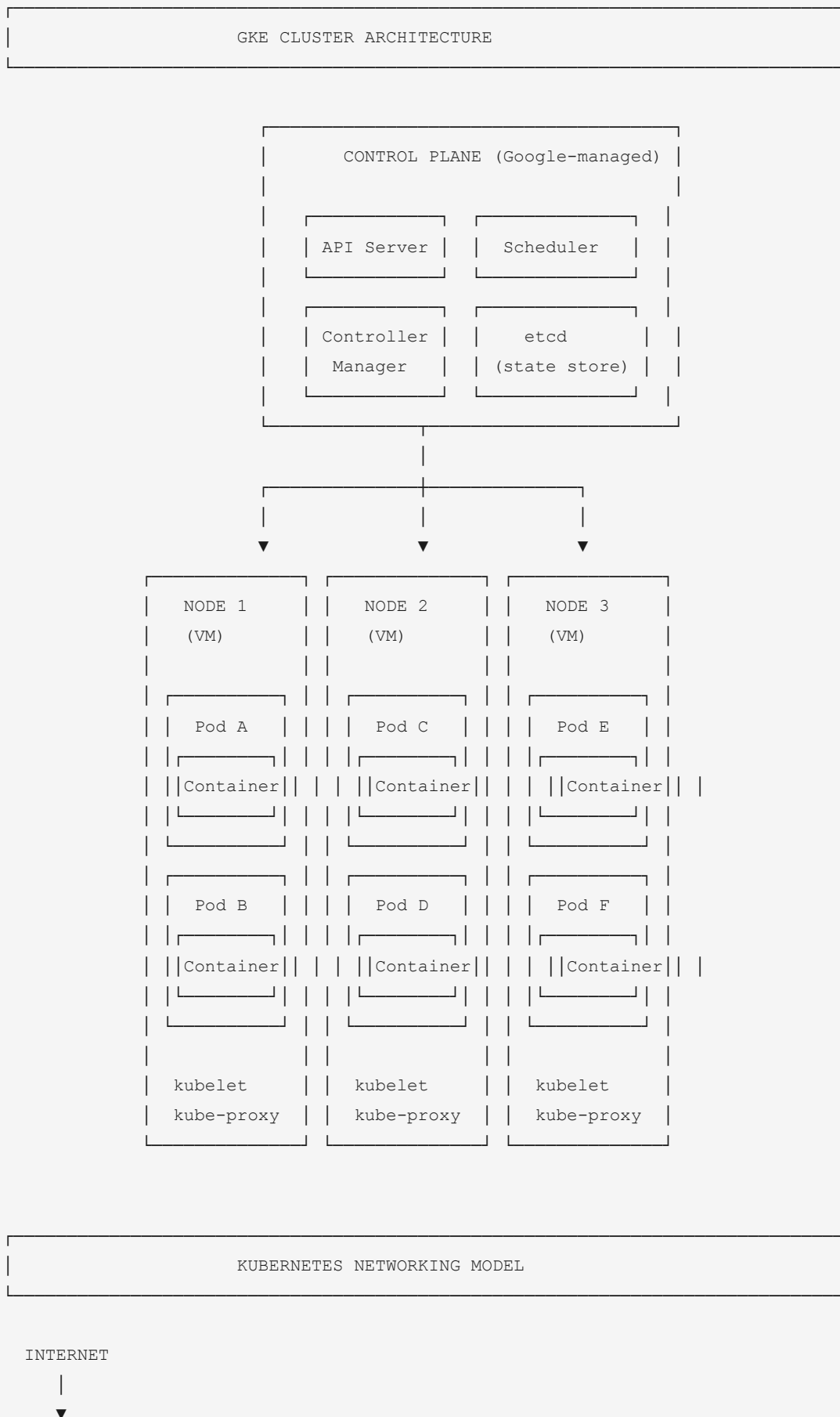
Study Notes April 2026

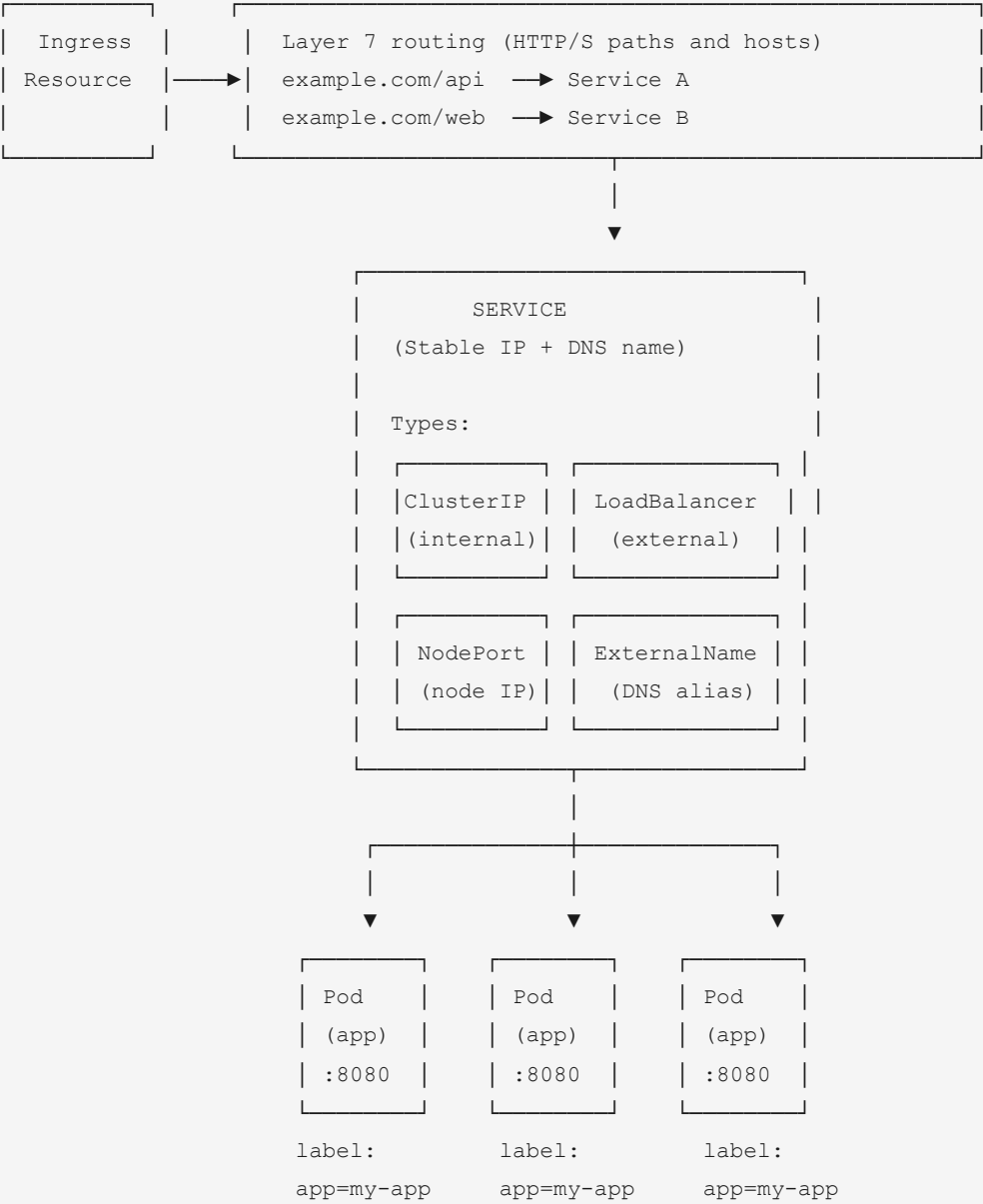
1. VM Deployment Flow: Template to Load Balancer



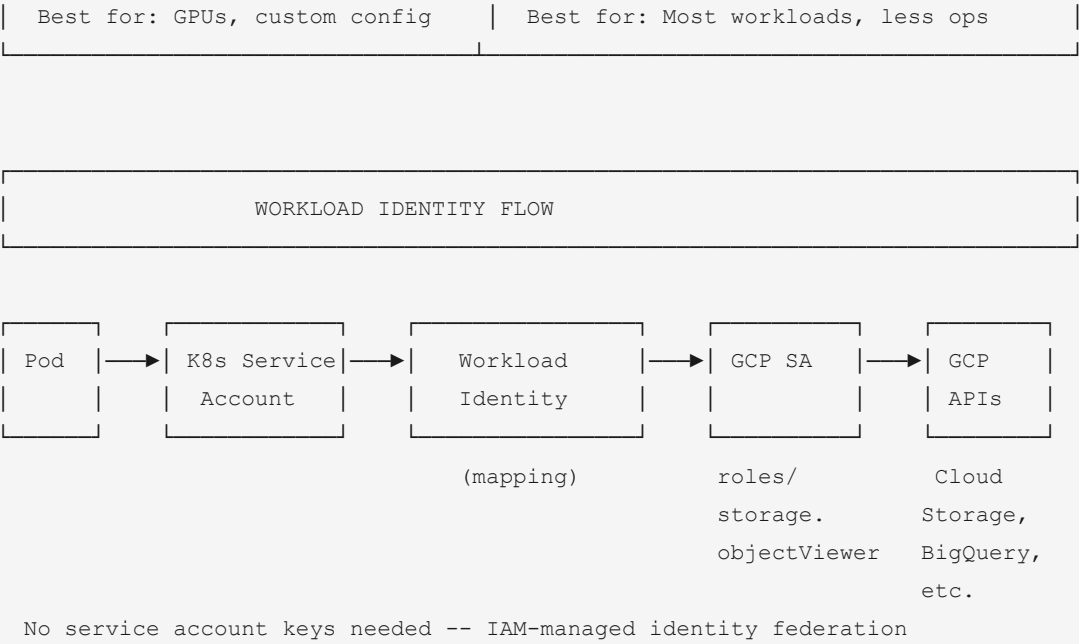


2. GKE Architecture: Control Plane to Services

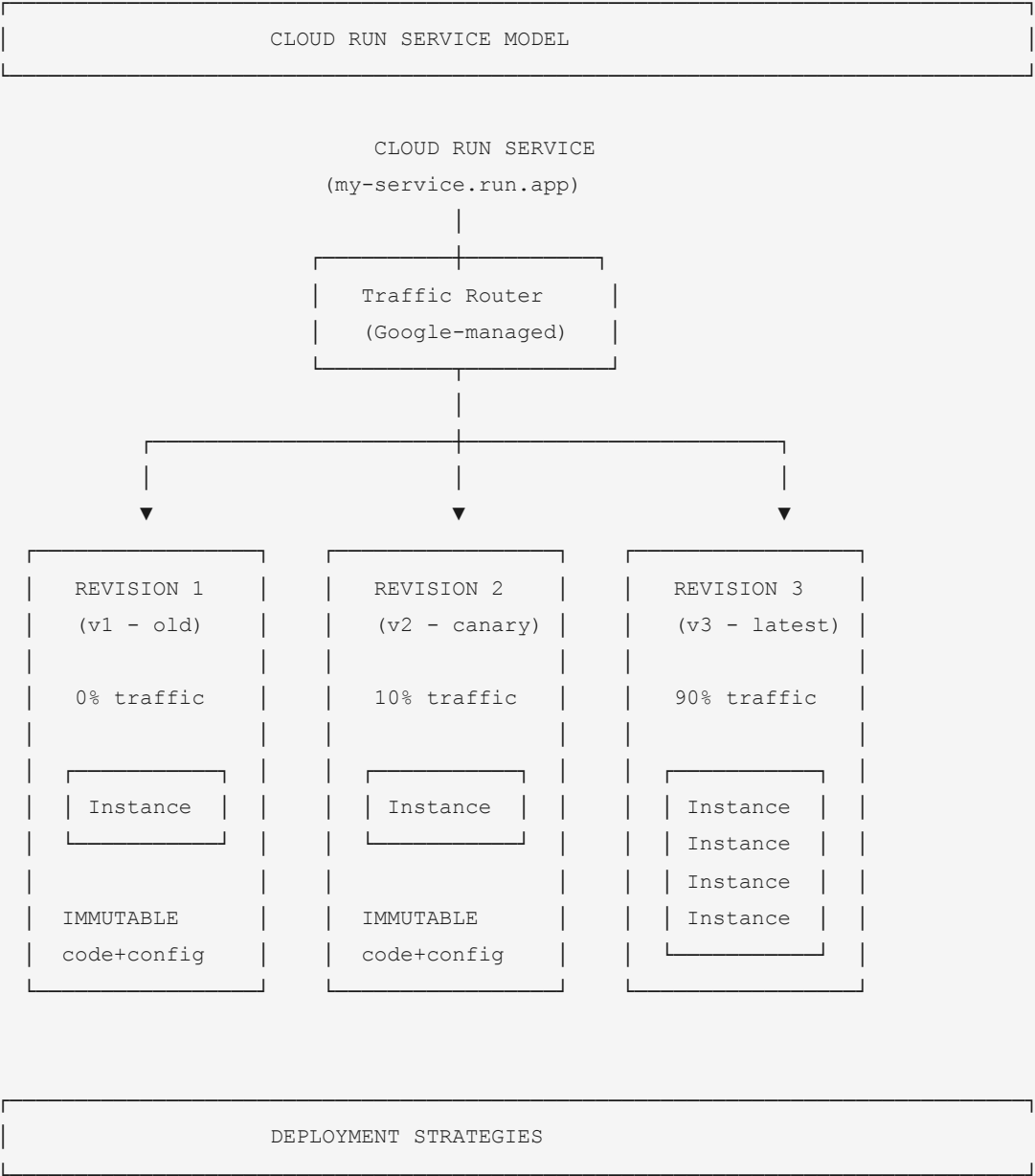




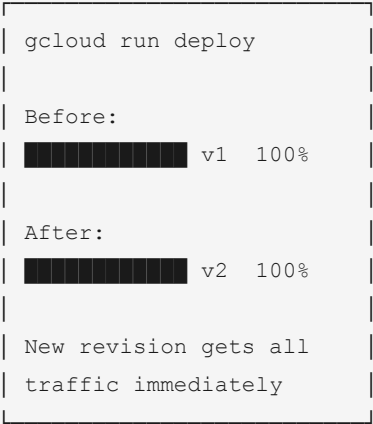
STANDARD vs AUTOPILOT CLUSTER	
STANDARD	AUTOPILOT
YOU manage: - Node pools - Machine types - Autoscaling config - Security hardening	GOOGLE manages: - Nodes - Security hardening - Scaling - Bin-packing
Pay for: NODE VMs	Pay for: POD resources (CPU/mem/disk)
Control: Full	Control: Limited
Overhead: Higher	Overhead: Lower



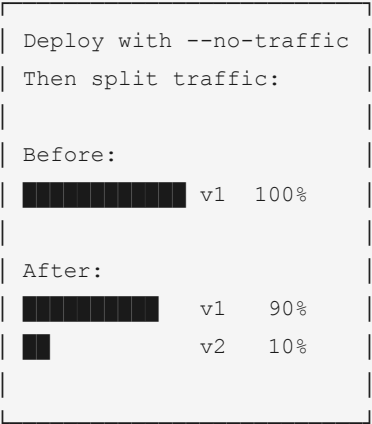
3. Cloud Run Revision and Traffic Model



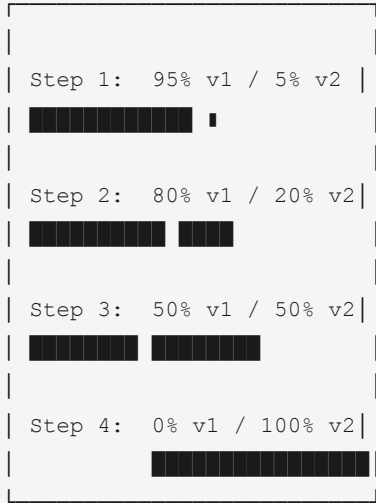
Strategy 1: INSTANT (default)



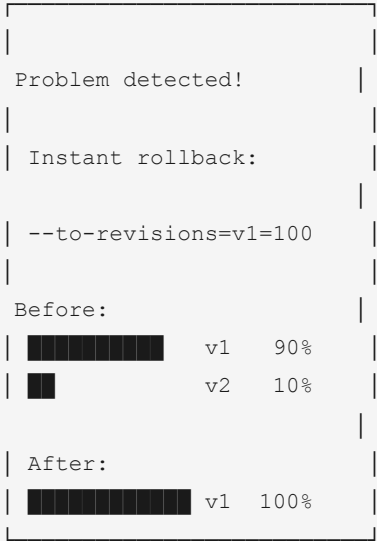
Strategy 2: CANARY



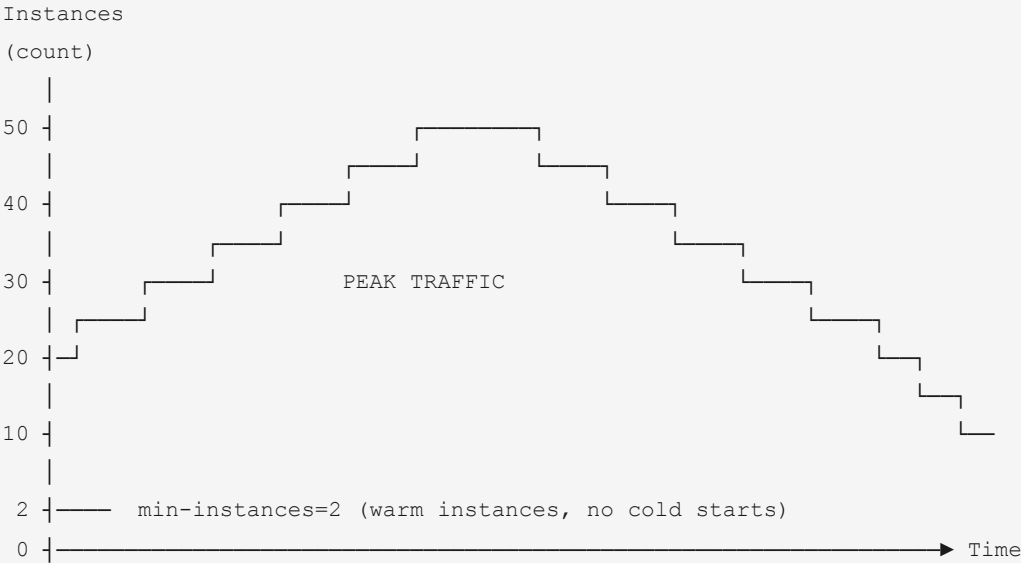
Strategy 3: GRADUAL ROLLOUT



Strategy 4: ROLLBACK



CLOUD RUN AUTOSCALING

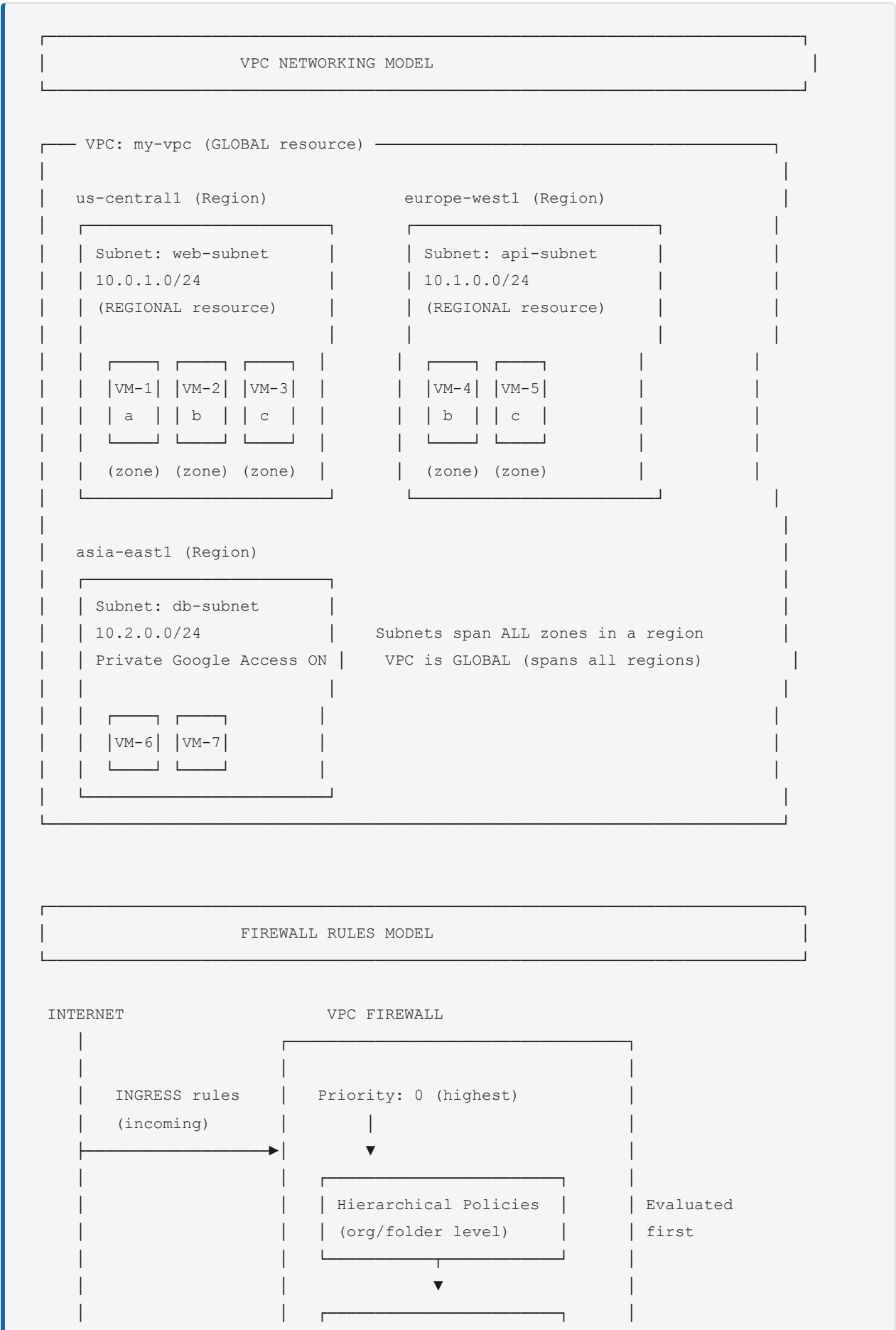


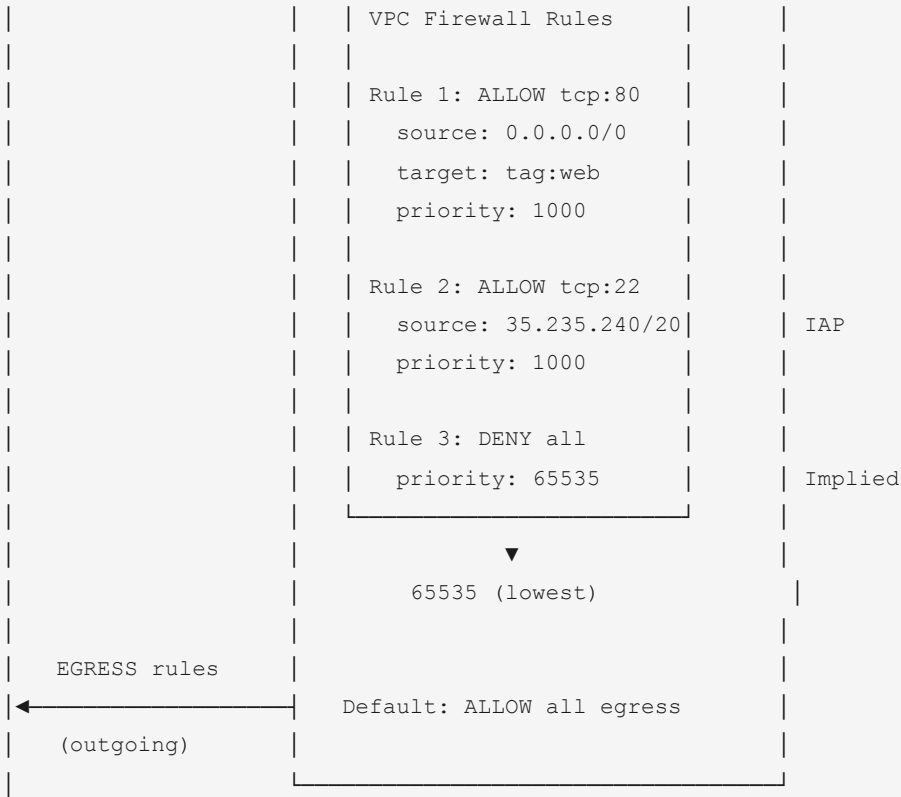
max-instances=50 (ceiling)
concurrency=80 (requests per instance)

Key parameters:

min-instances=0	Scale to zero when idle (cheapest, cold start)
min-instances=1	1 warm instance always (avoid cold starts)
min-instances=N	N warm instances (for latency SLAs)
concurrency	Higher = fewer instances = lower cost

4. VPC / Subnet / Firewall Visual Model

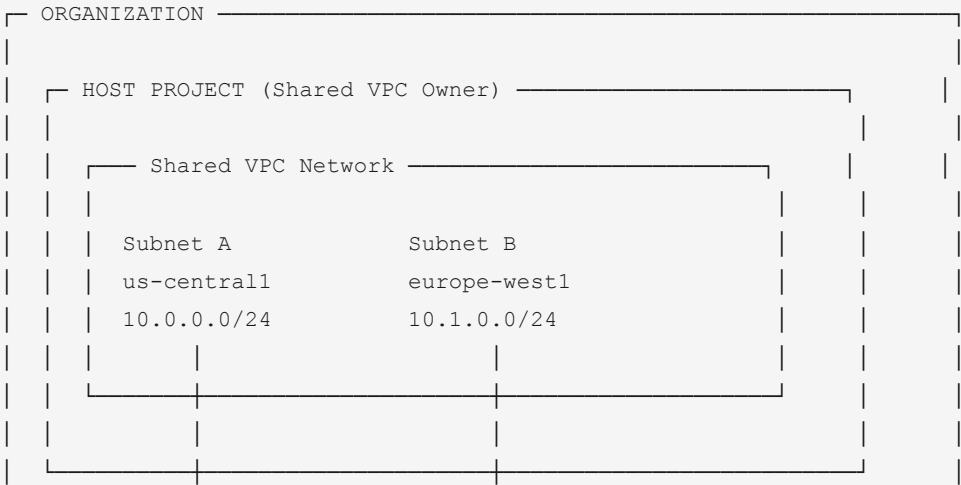


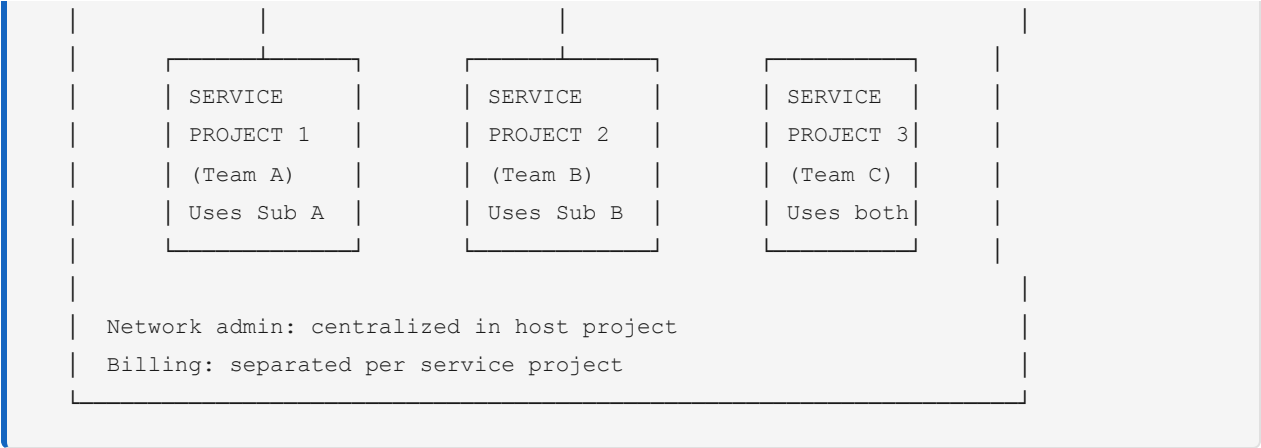


TARGETING METHODS:

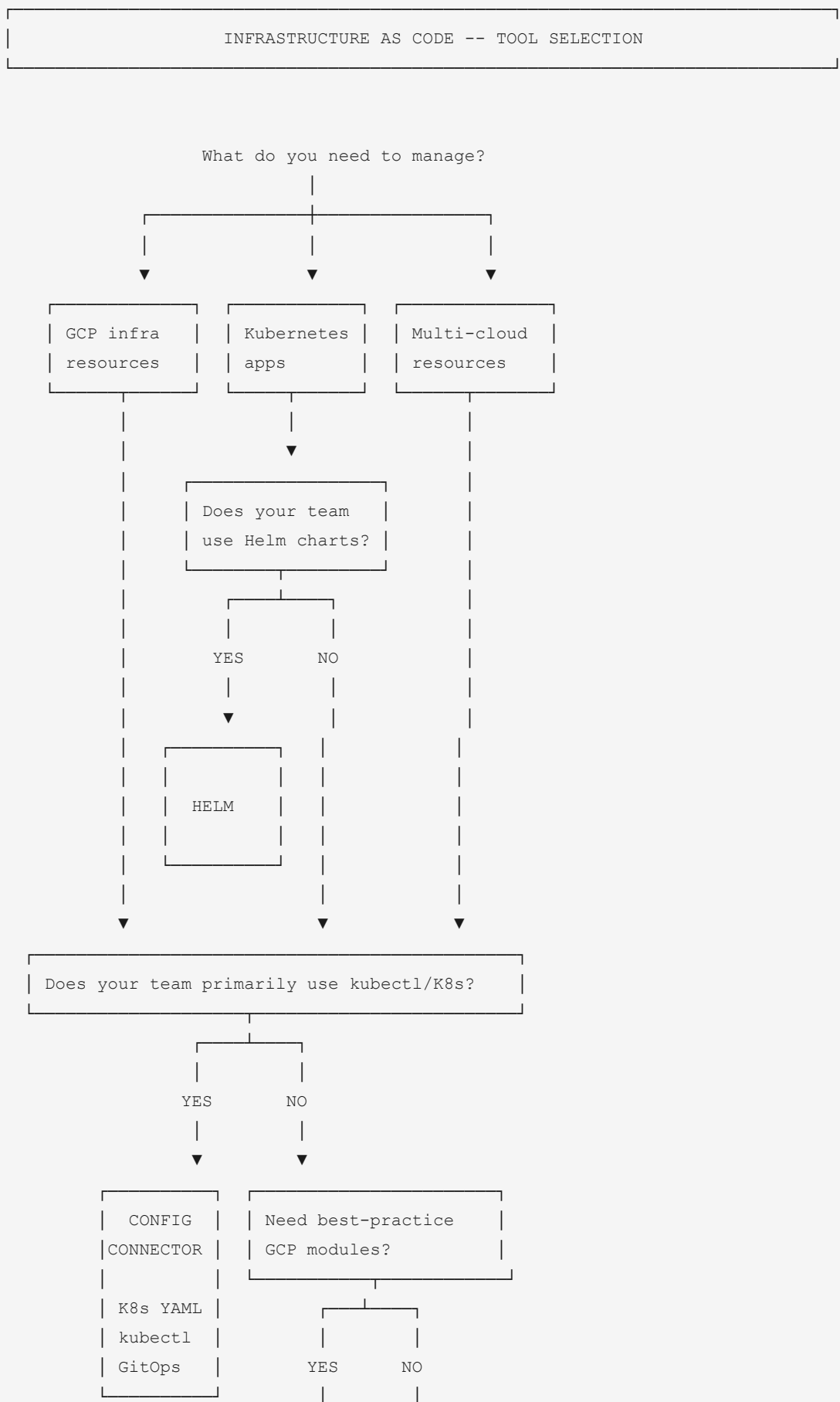
1. NETWORK TAGS	2. SERVICE ACCOUNTS	3. ALL INSTANCES
--target-tags=web	--target-service-accounts=sa@...	(no target = all)
Easy to manage	More secure	Broadest scope
Anyone can change	IAM-controlled	

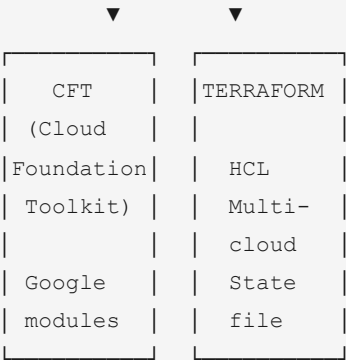
SHARED VPC MODEL





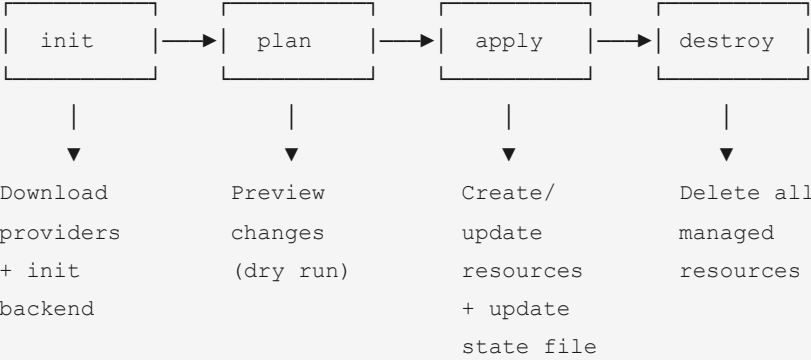
5. IaC Tool Selection Flowchart



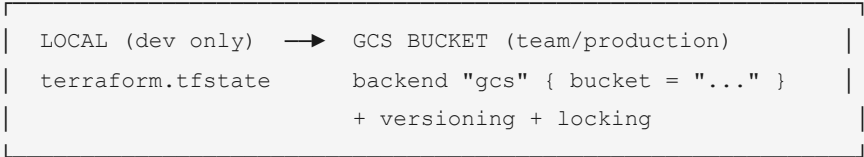


TOOL COMPARISON MATRIX				
	Terraform	CFT	Config Connector	Helm
Manages	Any cloud	GCP	GCP	K8s resources
Language	HCL	HCL	YAML	YAML + Go templates
Multi-cloud	YES	No	No	K8s only
Requires	CLI	CLI	GKE cluster	K8s cluster
State	File/GCS	File/GCS	etcd (K8s)	K8s API
Reconcile	On-demand	On-demand	Continuous	On-demand

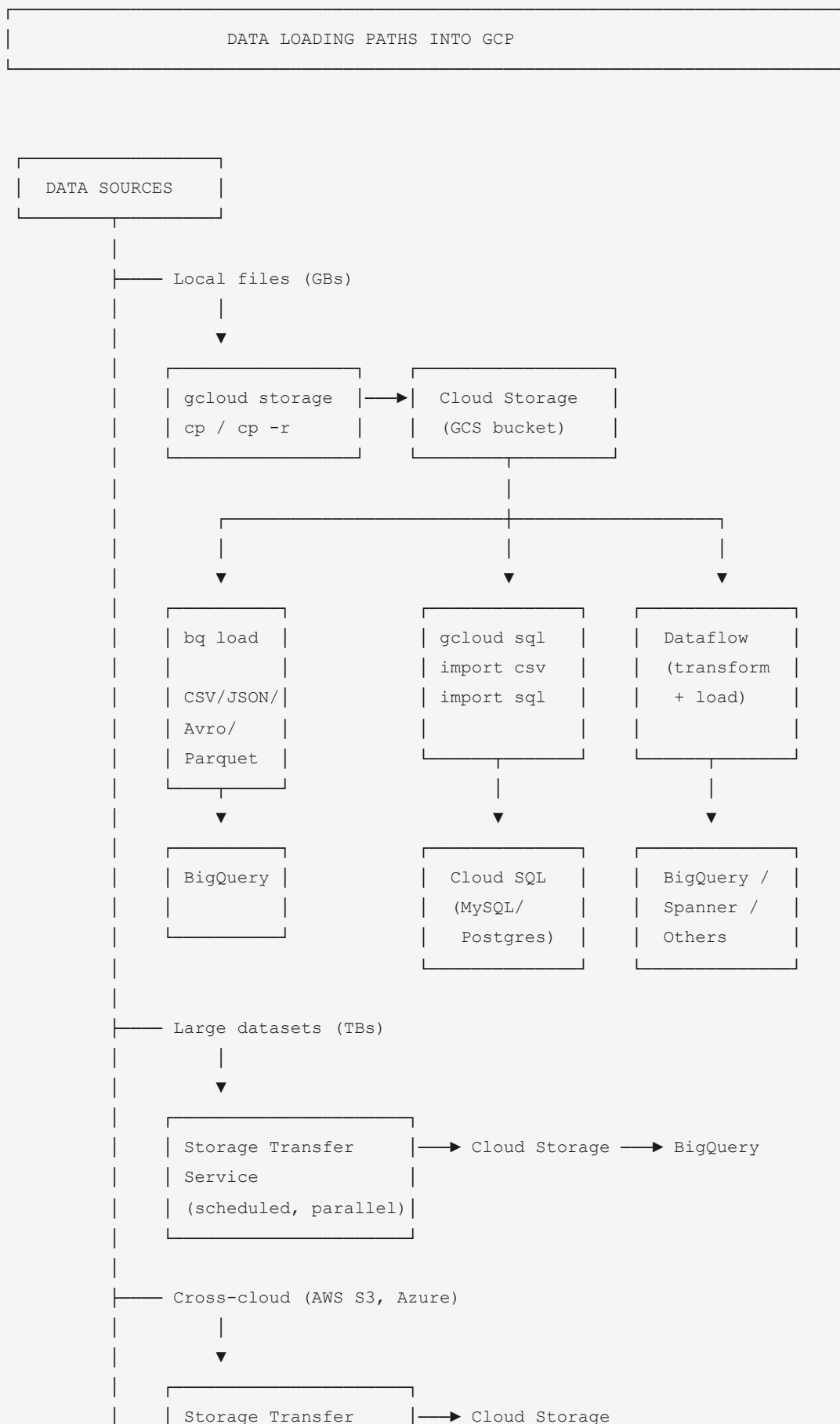
TERRAFORM WORKFLOW

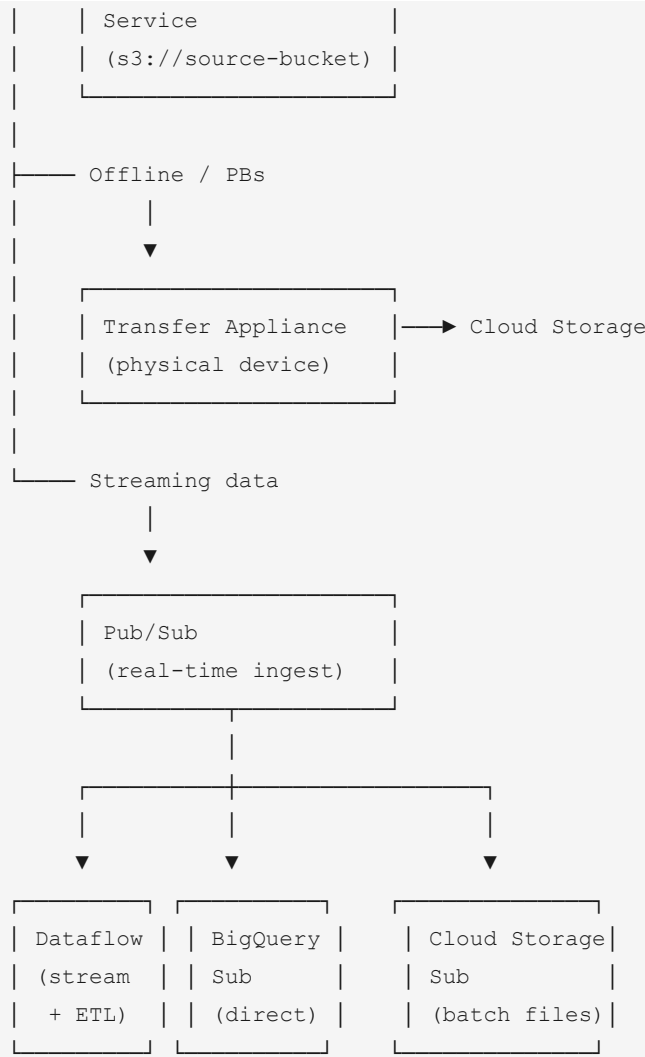


State Management:



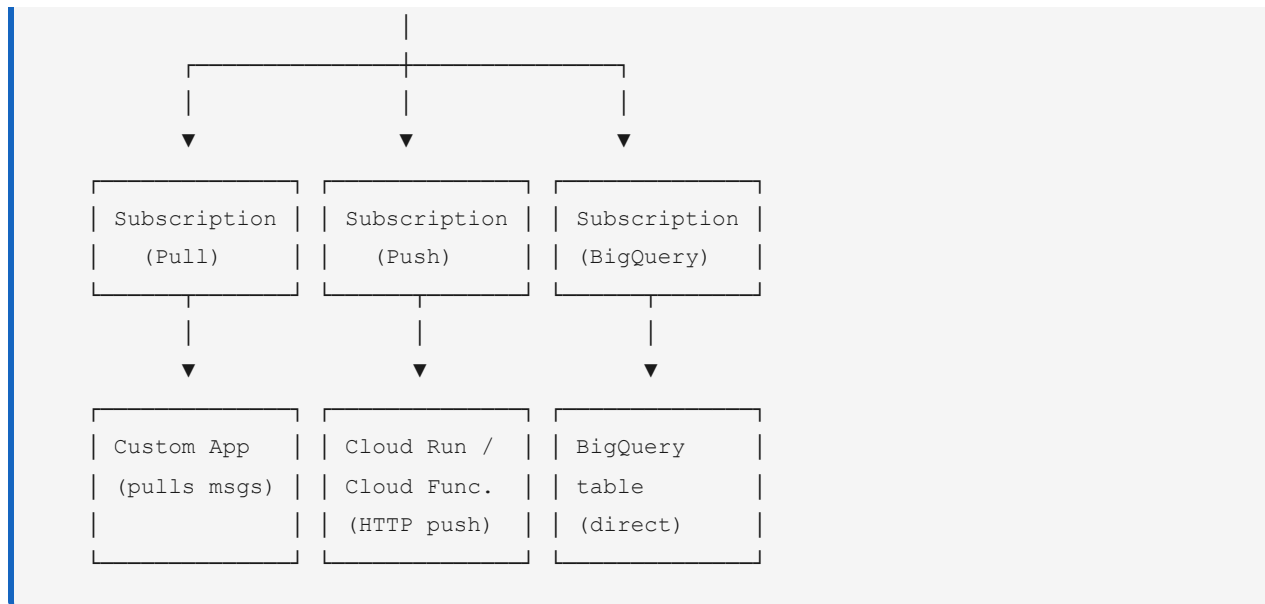
6. Data Loading Paths Diagram





TRANSFER METHOD SELECTION		
Method	Scale	Best For
gcloud storage	GBs	Ad-hoc uploads, small/medium files
bq load	GBs - TBs	Loading directly into BigQuery
Storage Transfer	TBs - PBs	Large-scale, scheduled, cross-cloud
Transfer Applian	PBs	Offline (ship physical device)
Dataflow	TBs	Transform-and-load ETL pipelines
Pub/Sub	Streaming	Real-time event/message ingestion

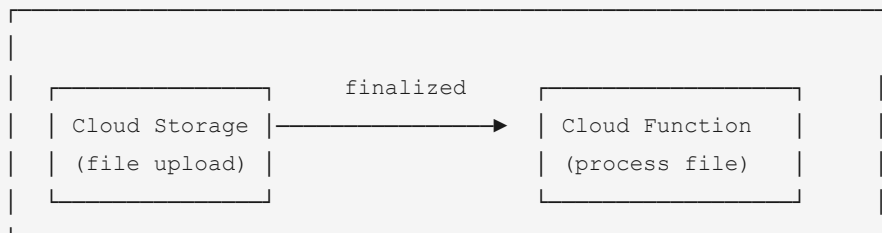




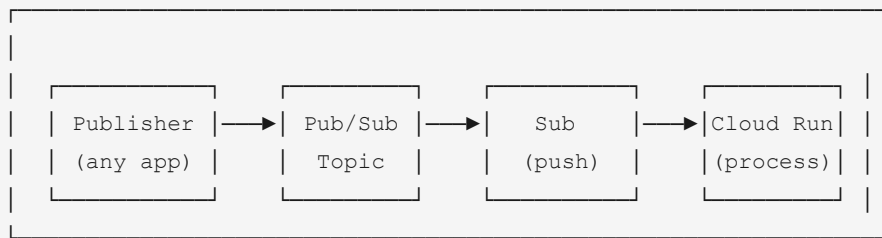
7. Event-Driven Architecture Patterns

EVENT-DRIVEN ARCHITECTURE PATTERNS

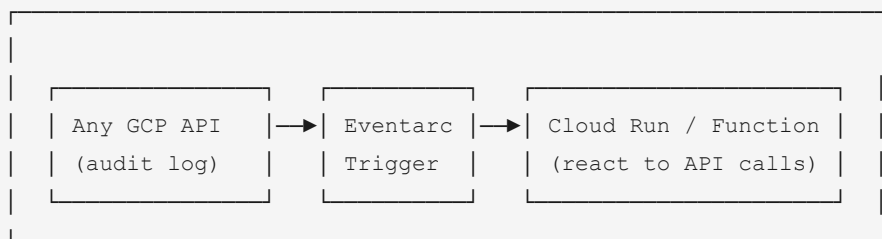
PATTERN 1: Direct Trigger



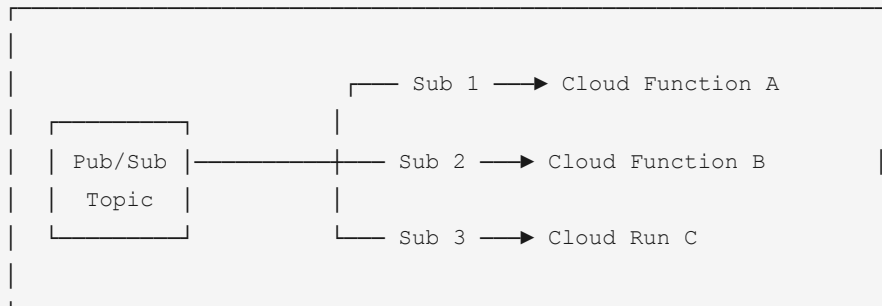
PATTERN 2: Pub/Sub Decoupling



PATTERN 3: Eventarc Routing (Audit Log Events)

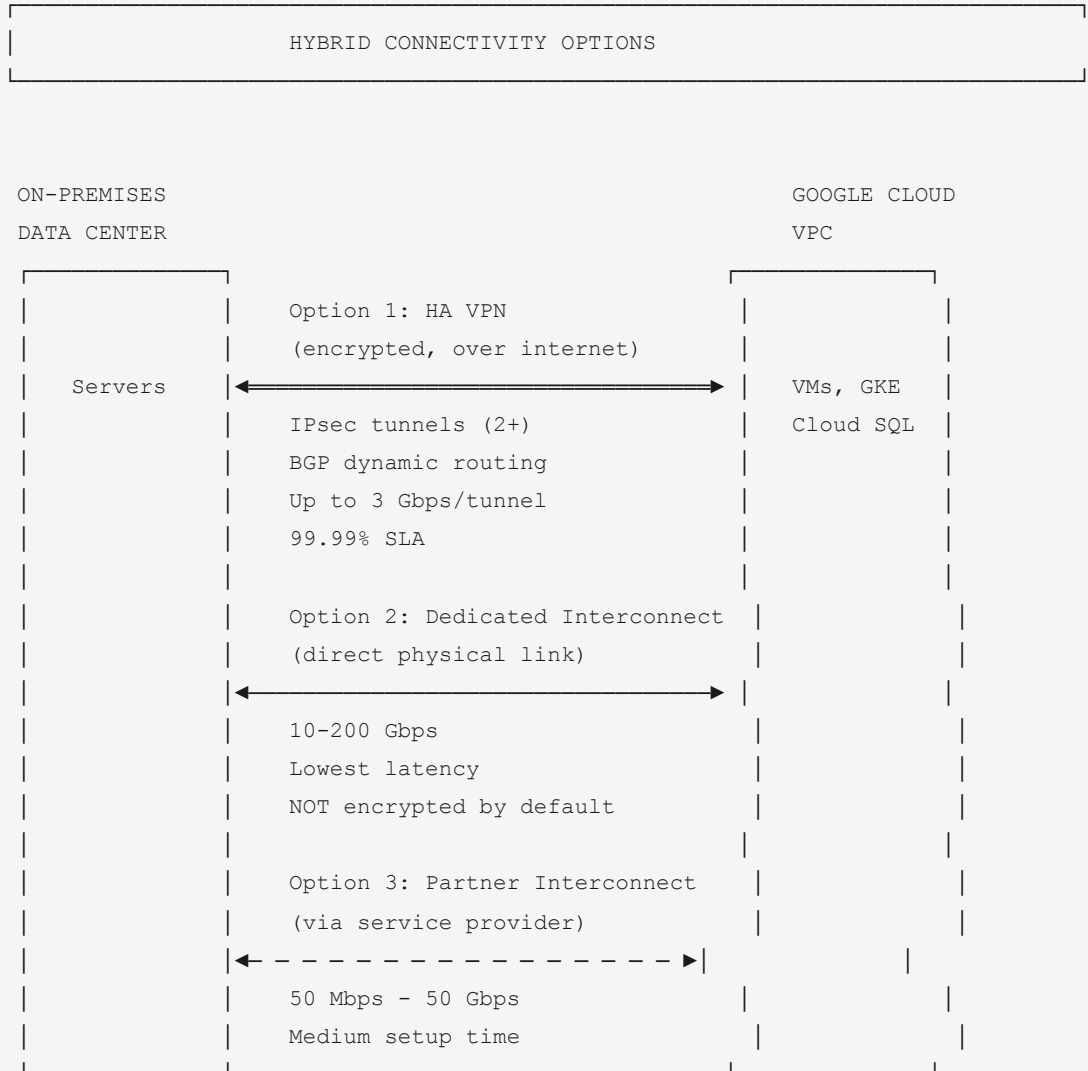


PATTERN 4: Fan-Out

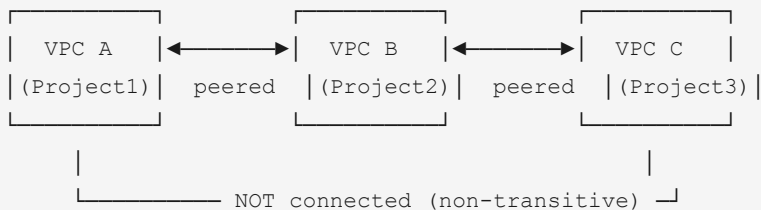


CLOUD RUN vs CLOUD FUNCTIONS -- DECISION GUIDE	
CLOUD RUN	CLOUD FUNCTIONS
Multiple endpoints/routes	Single-purpose function
Custom Dockerfile	Simple code in supported runtime
WebSocket / gRPC support	Quick event processing
Need > 32 GB memory	Short execution (< 60 min)
Traffic splitting	Simple Pub/Sub / Storage triggers
Migrating existing containers	Glue code between services
Team prefers containers	Team prefers writing functions

8. VPN and Network Connectivity



VPC PEERING:



Key: A can talk to B, B can talk to C, but A CANNOT talk to C through B
Both sides must configure peering. No overlapping IP ranges allowed.